1.2 Trigonometric Functions: The Unit Circle

Name _____ Period ____ Date _____

Pre-Calculus 1.2 · Student sheet

I can…

- find the point on the unit circle for a real number t.
- evaluate the six trigonometric functions using the unit circle.
- use the period and the even/odd properties to evaluate trigonometric functions.

Vocabulary · fill in the missing word

Unit circle — the circle $x^2 + y^2 = 1$, with _____ 1 and center at the origin

Sine and cosine — for the point (x, y) on the unit _____ for t: sin t = y and cos t = x

Periodic function — a function with f(t + c) = f(t) for all t; the _____ such c is the period

Part 1 · Special points

Hypotenuse = 1 (the radius).

t	π/6	π/4	π/3
Point (x, y)			

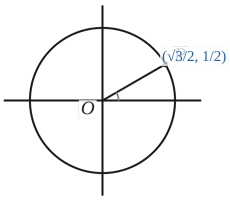
Part 2 · Notes and examples (fill in each note, then work the example)

For the point (x, y) for t: sin t = y, cos t = _____, tan t = y/x (x ≠ 0).

Reciprocals: csc t = 1/y, sec t = 1/x, cot t = _____.

Example 1

Evaluate all six trigonometric functions at $t = \frac{\pi}{6}$.



Example 2

Evaluate sin t, cos t, and tan t at $t = -\frac{\pi}{4}$.

Example 3

Evaluate all six functions at t = π.

Period: sin(t + 2πn) = sin t and cos(t + 2πn) = cos t, so sine and cosine have period _____.

Example 4

Evaluate.

a. $\sin \frac{13\pi}{6}$

b. $\cos(-\frac{7\pi}{3})$

Even and odd: cos(−t) = cos t (even); sin(−t) = _____ (odd).

Example 5

a. $\sin t = \frac{1}{3}$. Find sin(−t) and csc(−t).

b. $\cos t = \frac{3}{4}$. Find cos(−t) and sec(−t).

Part 3 · Partner practice

Partner 1

Evaluate sin t, cos t, and tan t at $t = \frac{5\pi}{3}$.

Partner 2

Evaluate $\cos \frac{9\pi}{4}$.